

Skill-guided agentic finite element analysis for nonlinear static verification of bridge catwalks

SCI manuscript framework | Adapted section and figure structure from FeaGPT | English draft outline

Reference structure

Qi, Y., Xu, R., and Chu, X. FeaGPT: an End-to-End agentic-AI for Finite Element Analysis. arXiv:2510.21993v1 (2025). <https://arxiv.org/abs/2510.21993v1>

Working title and outline follow the reference paper's end-to-end workflow, method modules, staged experiments and industrial demonstration. Engineering content is supplied by the catwalk Skill and the recorded six-case calculation.

Abstract: sentence plan

Context: automated verification of prestressed catwalk cable systems requires coordinated handling of model assets, load history, solver execution and engineering reports.

Method: introduce the English Skill, versioned inputs, identity-preserving Gmsh reconstruction, nonlinear CalculiX execution and automatic field/report generation.

Case: identify the Zhangjinggao Bridge catwalk model with 1,125 nodes, 1,194 elements and six dead/construction/wind/temperature combinations.

Results: insert 0.859887% maximum displacement difference, 0.853551% bottom-cable span-force difference and 1.125518% portal-cable span-force difference; report matching peak nodes and the calculated safety-factor outcomes.

Delivery: close with the reproducible solver evidence, 42 field plots and automatically formatted Word/PDF calculation report.

Keywords: agentic finite element analysis; executable skills; bridge catwalk; nonlinear static analysis; Gmsh; CalculiX; reproducibility.

I. Introduction

Reference anchor: I. Introduction, pp. 1–2

Paragraph 1 — Introduce agentic engineering workflows through the sequence used in FeaGPT: language-based task specification, tool execution and computational engineering results. Cite FeaGPT at the transition to finite element analysis.

Paragraph 2 — Move from the reference aircraft/component setting to bridge catwalk verification. Describe the practical input chain: a prestressed INP model, six load combinations, an existing engineering report and a required reproducible calculation package.

Paragraph 3 — Define the central workflow problem: preserving model identity, prestress, stepwise total loads and result-group definitions across automated execution. Use the P5 file-version/load-history episode as the motivating example.

Paragraph 4 — Introduce the Skill as the executable analysis specification, then list the paper contributions in the reference order: structured task execution; Gmsh–CalculiX integration; six-case quantitative verification; automatic engineering-report delivery.

II.A. System overview

Reference anchor: II.A. System Overview, p. 3; Fig. 1

Describe the four workflow modules and their file interfaces using Figure 1. The assistant reads the English Skill and invokes the packaged tools; each tool produces named, machine-readable evidence.

Identify the model asset manifest, original PDF, benchmark JSON and font/style assets. Trace one request from asset verification to the report and raw solver package.

II.B. Skill-guided planning and task orchestration

Reference anchor: II.B. Engineering Analysis Planning and Task Orchestration, pp. 3–6; Algorithm 1

Replace the reference geometry-design JSON with the catwalk task schema: input manifest, cases P1–P6, solver identity, units, execution stages, comparison criteria and report outputs.

Present Algorithm 1 as the executable order: validate > reconstruct mesh > solve each case > read final fields > recover forces > compare > generate report > hash outputs. State the stage-specific completion conditions.

For the planned GLM execution experiment, record provider/model identifier, request, tool calls, elapsed time, retries and final artifact status in a session log. Fill the model identifier from the actual run.

II.C. Versioned model reconstruction and engineering knowledge

Reference anchor: II.C. Intelligent Geometry Generation with Knowledge Augmentation, pp. 6–7; Fig. 2

Describe reconstruction from the original coordinates and connectivity, followed by material/section/support audit. Use Figure 2 to connect global geometry, structural families and a local connection.

Give the equivalent material values from the INP: MAT1/MAT2 E = 120 GPa, areas 22298.692/8402.9797 mm² and their equivalent densities; MAT3 E = 206 GPa, rectangular section 98.954535 × 98.954535 mm and near-zero assigned density.

Describe 8,984 initial-stress records across 1,123 cable elements, the global tensor convention and the retained reference geometry. Link numerical values to the material table and asset hashes.

II.D. Gmsh mesh generation and identity mapping

Reference anchor: II.D. Adaptive Mesh Generation, pp. 7–8; Fig. 3; Table II

Explain point/line creation from INP IDs, section physical groups, setTransfiniteCurve(eid, 2), first-order generate(1), and MSH 4.1 export.

Distinguish geometric tags from mesh tags. Describe the explicit node/element mapping and checks: node count, element count, coordinates and ordered endpoints. Table II reports the measured mesh identity results.

II.E. Nonlinear static analysis and load-history execution

Reference anchor: II.E. Automated FEA Simulation and Analysis, pp. 8–9

Write equilibrium as $R(u, \lambda) = \text{fint}(u, \sigma_0, T) - \text{fext}(\lambda) = 0$. Describe geometric nonlinearity, initial prestress, gravity and prescribed supports within the planar X–Z system.

State the six combinations in Table I. P1 contains one dead-load step; P2–P6 independently establish dead load before their target total combination. Explain the second-step CLOAD total convention, with P5 as the worked example.

Give $g = 9806 \text{ mm/s}^2$, $\alpha = 1.2\text{e-}5 \text{ /}^\circ\text{C}$ for P3/P6, temperature changes -15/-34°C and the 1,125 UY constraints. Identify the locked solver build and final-time checks.

II.F. Field recovery and automated verification

Reference anchor: II.F. Intelligent Data Analysis and Batch Processing, p. 9

Define $U_{\max} = \max \sqrt{U_X^2 + U_Y^2 + U_Z^2}$ over original nodes. Define the deformed chord n and recover $N = A_0 n^T \text{sigmabar } n / 1000$ in kN, averaging the eight integration-point tensors.

Define eight reporting span groups, their maxima and $K = N_{\text{break}}/N_{\max}$. Set $N_{\text{break}} = 38080/14280 \text{ kN}$ and give the case-specific K requirements.

Define signed relative difference $\delta = 100(q_{\text{calc}}/q_{\text{ref}} - 1)$. Identify the independent MAPDL displacement source and the report force tables separately. Describe Figure 4 and the machine-readable evidence files.

II.G. Reproducible batch execution and report generation

Reference anchor: II.G. Scalable Batch Processing Architecture, pp. 9–10

Describe independent case directories, common geometry audit, recorded solver commands, failure preservation and per-stage status.

Trace one final response value into CSV, VTU, PNG and the engineering report. Describe the shared result model used for Word/PDF and the style/font assets reproducing the reference report.

State the deliverable structure: six solved cases, 42 field plots, comparison data, Gmsh mapping, hashes and a 17-page engineering report in both formats.

III.A. Task specification and six-case execution

Reference anchor: III.A. Natural Language Input and System Understanding, p. 11; Table I

Present the task request and its mapped case/output specification. Table I enumerates the six combinations and required safety factors.

For the GLM session, insert the recorded model ID and stage-completion log. Report execution and retry counts from that log.

III.B. Model reconstruction results

Reference anchor: III.B. Geometry Generation Results, pp. 11–12; Fig. 2

Report 1,125 original nodes, 1,123 T3D2 cable elements and 71 B31 equivalent members. Show the complete geometry and member assignments in Figure 2.

Report the common geometric/physical definitions across cases and the preserved initial-stress record count.

III.C. Mesh identity and mapping results

Reference anchor: III.C. Mesh Generation and Quality Analysis, pp. 12–13; Table II; Fig. 3

Report identical node/element counts, zero coordinate discrepancy and matching ordered connectivity between the input and the Gmsh mesh. Use the local inset to display the mapping.

Explain how this identity connects solved fields to the displayed geometry and reporting groups.

III.D. Nonlinear response and benchmark comparison

Reference anchor: III.D. FEA Simulation and Results Analysis, p. 14; Table III

Report completion of all six cases and matching peak-displacement node IDs. Present the six numerical rows in Table III.

Give maximum absolute differences: displacement 0.859887%, bottom-cable span force 0.853551% and portal-cable span force 1.125518%. Identify the controlling case/span from the supplied evidence table.

Discuss P4 as the largest peak-displacement response, P5 peak at node 306, and the P3/P6 cooling responses using the solved fields.

III.E. Cable response interpretation and safety assessment

Reference anchor: III.E. Intelligent Result Analysis and Autonomous Design Evaluation, pp. 14–16; Fig. 4

Follow the four panels in order: global displacement agreement, span-force agreement, cable safety factors and the spatial/case distribution of signed differences.

Compare the governing bottom/portal factors by case with Table I requirements. Use the heatmap to locate differences while retaining cable family and span identity.

Explain the load-history correction through the P5 input audit: second-step total nodal load includes the first-step permanent contribution. Connect the input definition with the current node-306 downward response.

III.F. Execution performance and automation accounting

Reference anchor: III.F. Computational Performance and Scalability, pp. 16–17

Report the recorded solver wall times by case and their sum, with Python/platform/solver identity. Label timing boundaries explicitly.

Add the planned GLM session measurements in a separate table: planning time, tool time, report-generation time, total session time, token usage when exposed by the provider, retries and completion count.

For repeated-session evaluation, use the same six inputs and output contract; record each session independently before calculating completion rate and timing distributions.

III.G. Engineering case presentation and report delivery

Reference anchor: III.G. Industrial Validation: CalculiX Configuration Generation, pp. 17–18; Fig. 5

Present Figure 5 as the engineering field plate: three representative cases and two response quantities. Compare shared field scales within each row.

Show how the prescribed report style is generated from the same result set, and point to the supplementary six-case contour plates, Word/PDF report and raw result archive.

IV. Conclusions

Reference anchor: IV. Conclusions, pp. 18–19

Conclusion 1 — Summarize the Skill-guided sequence connecting versioned bridge inputs to mesh verification, nonlinear solution and engineering-report generation.

Conclusion 2 — State the measured mesh identity and six-case response differences with their actual values.

Conclusion 3 — State the cable safety-factor outcomes and delivered evidence package. Add session-level GLM findings after inserting the measured session table.

Table I. Case specification

Case	Combination	Steps	Delta T / °C	Required K
P1	Dead load	1	0	3.2
P2	Dead + construction	2	0	2.7
P3	Dead + construction + cooling	2	-15	2.7
P4	Dead + construction + wind	2	0	2.7
P5	Dead + maximum gust	2	0	2.5
P6	Dead + construction + cooling	2	-34	2.7

Table II. Mesh and model identity

Metric	Recorded value
Original / Gmsh nodes	1125 / 1125
Original / Gmsh line elements	1194 / 1194
Structural element families	1123 T3D2 + 71 B31
Coordinate discrepancy	0 mm
Ordered connectivity	All 1194 elements match
Initial-stress records	8984
Reporting span groups	4 bottom + 4 portal
Reported-span cable elements	1113
Original-node planar constraints	1125 UY constraints

Table III. Six-case response and solver timing

Case	Umax/mm	Node	delta U/%	K bottom	K portal	Solver/s
P1	42.259	304	-0.7494	3.3165	4.0135	0.768
P2	2662.041	1176	-0.8491	3.0165	3.5671	1.777
P3	1897.223	1176	-0.6696	3.0056	3.5574	1.822
P4	2846.280	1176	-0.8599	2.9895	3.5268	1.870
P5	1325.314	306	-0.7830	3.1873	3.8802	1.724
P6	927.260	1166	-0.0166	2.9464	3.4677	1.672

Recorded solver wall time summed across six sequential cases: 9.633834 s. Timing source: examples/run.json, executions[].elapsed_s; timing covers the solver subprocess. Run ID: 20261006T174516Z.

Algorithm 1. Skill-guided execution

1. Read task and resolve the versioned six-case manifest.
2. Verify asset hashes and audit material, section, prestress, load and support cards.
3. Regenerate the Gmsh line mesh and verify original-to-mesh identity.

4. For each case P1–P6: copy its complete INP into a new case directory and execute the recorded solver command.
5. Validate final step/time and complete finite U/S fields.
6. Recover deformed-chord cable forces; evaluate span maxima, safety factors and benchmark differences.
7. Write CSV, VTU and seven field plots per case.
8. Generate the Chinese engineering report from the common result object and prescribed style assets.
9. Write execution status and output hashes; return the report and evidence links.

Planned GLM session table

Field	Value source
Provider/model/version	Actual GLM session metadata
Request and tool trace	Saved request and tool-call log
Planning / tools / report / total time	Session stage timestamps
Token usage	Provider usage record
Retries and completed stages	Execution log
Repeated-session completion rate	Completed sessions / attempted sessions

Figure list and captions

Figure 1

Figure 1. Skill-guided workflow for nonlinear static verification of the catwalk cable system, connecting versioned inputs, mesh verification, finite element solution, field recovery and engineering-report generation.

Panel construction: Keep the left-to-right request-to-result composition and the knowledge block above the workflow. Replace the aircraft request with the catwalk INP set and reference report. Use four module boxes: input/model audit; Gmsh topology reconstruction; nonlinear CalculiX execution; field recovery and report generation. Add the manifest, Skill and benchmark assets above, and the Word/PDF/CSV/VTU outputs at the return arrow.

Figure 2

Figure 2. Catwalk finite element model: (a) longitudinal profile and control points; (b) cable families and equivalent members; (c) a portal connection with element assignments and boundary conditions.

Panel construction: Retain three panels and the global-to-local zoom. Panel (a): full X–Z catwalk profile and control points. Panel (b): bottom cables, portal cables and equivalent members in three colors. Panel (c): enlarged portal/cable connection with original node and element IDs, material assignments and supports. Draw geometry directly from the INP coordinates.

Figure 3

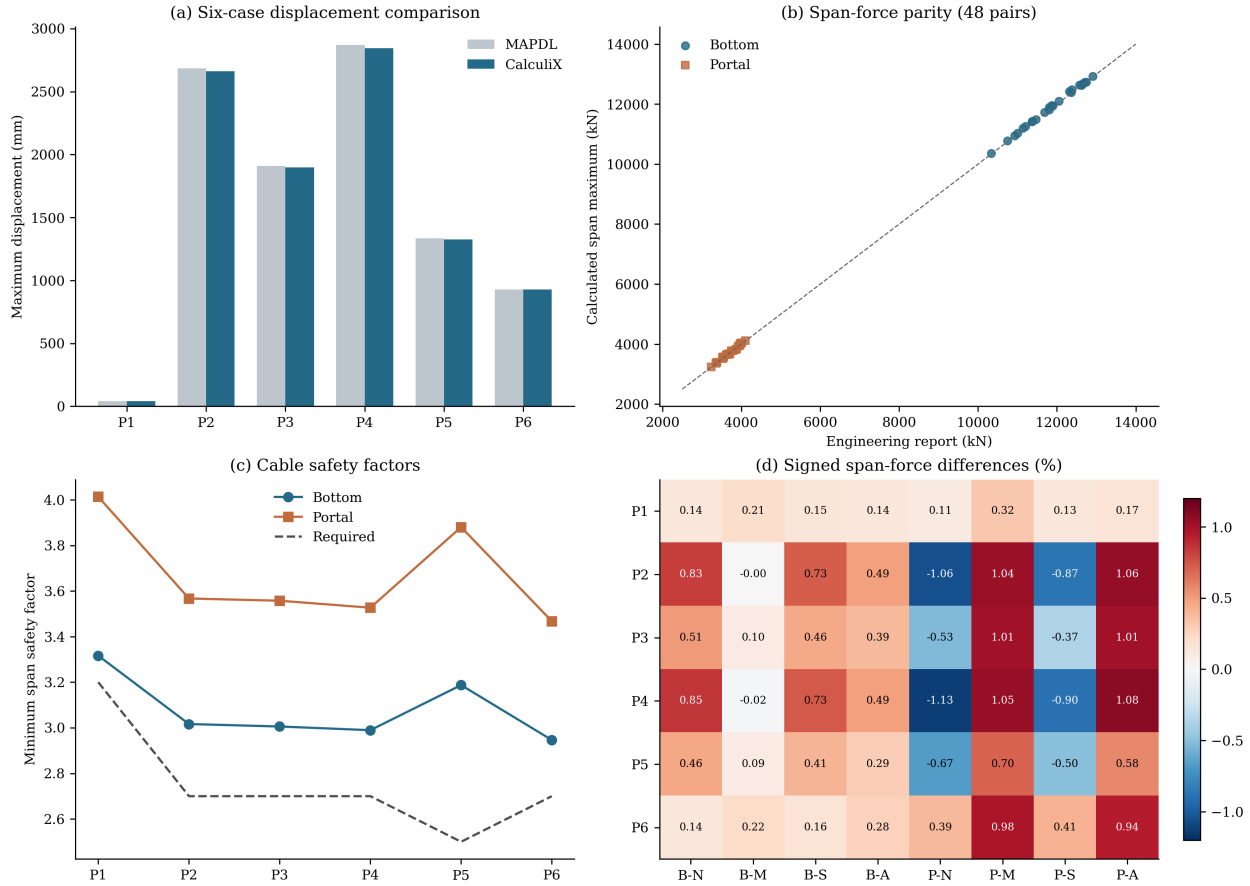
Figure 3. Mesh reconstruction and field transfer: (a) Gmsh mesh and original-input tag correspondence; (b) P5 bottom-cable axial-force distribution on the verified line mesh.

Panel construction: Retain the two main panels and inset. Panel (a): original/Gmsh line mesh overlay with an enlarged tag-mapping detail. Annotate 1,125 nodes, 1,194 elements and zero coordinate discrepancy. Panel (b): P5 axial-force line contour using the same geometry; show kN, span group and governing element. Draw only line elements.

Figure 4

Figure 4. Six-case verification: (a) maximum displacement compared with MAPDL; (b) 48 span-force comparisons with the engineering report; (c) minimum span safety factors and required values; (d) signed force differences by case and span.

Panel construction: Keep the 2×2 evidence layout. Panel (a): six-case CalculiX/MAPDL peak-displacement bars. Panel (b): parity scatter for 48 report/calculated span-force pairs, distinguished by cable family. Panel (c): minimum four-span safety factor by family and case with required values. Panel (d): signed relative-error heatmap for six cases × eight span groups. Use the actual summary JSON for every point.



B/P = bottom/portal cables; N/M/S/A = north side/main/south side/south auxiliary span.

Figure 5

Figure 5. Representative field responses under dead load, maximum gust and cooling: upper row, vertical displacement; lower row, bottom-cable axial force. Columns correspond to P1, P5 and P6.

Panel construction: Keep six panels, arranged as three columns × two rows. Columns: P1 dead load, P5 maximum gust and P6 cooling. Upper row: signed UZ. Lower row: bottom-cable axial force. Use common limits within each field row, label mm/kN, and annotate the relevant extreme node/element. Place all six load cases in supplementary field plates using the same settings.

Supplementary material

S1: English Skill, seven execution references, source scripts and environment requirements.

S2: Input manifest, asset hashes, material/load audits and Gmsh mapping.

S3: Complete 48-row span-force comparison and six-case displacement table.

S4: Six-case solver logs, DAT/FRD/STA, CSV/VTU and 42 field plots.

S5: Generated Word/PDF engineering report and typography specification.

S6: GLM session trace and repeated-session measurements, populated from the planned execution experiment.

References and data provenance

Qi, Y., Xu, R., and Chu, X. FeaGPT: an End-to-End agentic-AI for Finite Element Analysis.
arXiv:2510.21993v1 (2025). <https://arxiv.org/abs/2510.21993v1>

Zhangjinggao Yangtze River Bridge, South Navigation Channel Bridge: Catwalk Structural Review Calculation Report, version 0324, 118 pages. Project-supplied PDF; force benchmarks from Tables 1-11 and 1-14, safety-factor criteria from Table 1-7.

Catwalk static review Skill v1.2.0. Numerical evidence: examples/summary.json, examples/comparison.csv and examples/run.json. Input, solver and PDF identities are recorded in the package manifests.